

MICHAEL (mike) NEUDER

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🌐 Princeton, New Jersey, USA

EDUCATION

Princeton University <i>Doctoral student, Computer Science</i> ↳ Under the supervision of Prof. Matt Weinberg ↳ Supported by an Ethereum Foundation PhD Fellowship (FY25-2276)	Aug. 2025 - Present Princeton, NJ
Harvard University <i>Master of Science, Computational Science</i>	Aug. 2020 - May. 2021 Cambridge, MA
University of Colorado <i>Bachelor of Arts, Computer Science</i> <i>Bachelor of Arts, Mathematics</i>	Aug. 2015 - May. 2020 Boulder, CO

ACADEMIC EXPERIENCE

EconCS Group, Harvard University <i>Research Assistant supervised by Prof. David C. Parkes</i>	Aug. 2019 - Aug. 2021 Cambridge, MA
Santa Fe Institute <i>Undergraduate Research Fellow: Summer 2018 REU</i>	Jun. 2018 - Aug. 2018 Santa Fe, NM
Bradley Lab, University of Colorado <i>Research Assistant supervised by Prof. Elizabeth Bradley</i>	Apr. 2017 - Aug. 2021 Boulder, CO
Mozer Lab, University of Colorado <i>Research Assistant supervised by Prof. Michael Mozer</i>	Mar. 2017 - May. 2019 Boulder, CO

INDUSTRY EXPERIENCE

The Ethereum Foundation <i>Researcher – Applied Research Group (ARG)</i>	Mar. 2023 - May. 2025 New York, NY
Google <i>Software Engineer (L3 → L4) – Cloud Storage</i> <i>Software Engineering Intern – Network Infrastructure</i> <i>Software Engineering Intern – Flights</i> <i>Software Engineering Intern – Mobile Device Management</i>	Aug. 2021 - Feb. 2023 (Cambridge, MA) Summer 2020 (remote) Summer 2019 (Cambridge, MA) Fall 2018 (Sunnyvale, CA)
Lockheed Martin & LASP¹ <i>Software Engineering Intern</i>	Feb. 2017 - Oct. 2017 Boulder, CO

JOURNAL AND CONFERENCE PUBLICATIONS (REVERSE CHRONOLOGICAL)

- P. Garimidi, **M. Neuder**, T. Roughgarden, “Beyond Winner-Take-All Procurement Auctions,” (2026) *Financial Cryptography and Data Security (FC’26)*. <https://arxiv.org/pdf/2603.27779>.
- M. Bahrani, **M. Neuder**, S. M. Weinberg, “Selfish mining under general stochastic rewards,” (2025) *Advances in Financial Technologies (AFT’25)*.
<https://doi.org/10.4230/LIPIcs.AFT.2025.20>. (full version on arxiv: <https://arxiv.org/abs/2502.20360>.)

¹Laboratory of Atmospheric and Space Physics: <https://lasp.colorado.edu/home/>.

- **M. Neuder**, M. Pai, and M. Resnick, “Optimizing Exit Queues for Proof-of-Stake Blockchains: A Mechanism Design Approach,” (2024) *Advances in Financial Technologies (AFT’24)*.
<https://doi.org/10.4230/LIPIcs.AFT.2024.20>.
- Z. Fan, F. Marmolejo-Cossio, D. J. Moroz, **M. Neuder**, R. Rao, and D. C. Parkes, “Strategic Liquidity Provision in Uniswap v3,” (2023) *Advances in Financial Technologies (AFT’23)*.
<https://doi.org/10.4230/LIPIcs.AFT.2023.25>.
- **M. Neuder**, E. Bradley, E. Dlugokencky, J. W. C. White, J. Garland, “Detection of Local Mixing in Time Series using Permutation Entropy,” (2021) *Physical Review E* 103.
<https://doi.org/10.1103/PhysRevE.103.022217>.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Defending Against Malicious Reorgs in Tezos Proof-of-Stake,” (2020) *ACM Conference on Advances in Financial Technologies (AFT’20)*.
<https://doi.org/10.1145/3419614.3423265>.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Selfish Behavior in the Tezos Proof-of-Stake Protocol,” (2020) *Cryptoeconomic Systems (CES’20) Conference*. <https://arxiv.org/pdf/1912.02954>.
- J. Garland, T. Jones, **M. Neuder**, J. W. C. White, E. Bradley, “An information-theoretic approach to extracting climate signals from deep polar ice cores,” (2019) *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29:101105. <https://doi.org/10.1063/1.5127211>.
- J. Garland, T. Jones, **M. Neuder**, V. Morris, J. W. C. White, E. Bradley, “Anomaly Detection in Paleoclimate Records using Information Theory,” (2018) *Entropy* 20(12):931. <https://doi.org/10.3390/e20120931>.

WORKSHOP PAPERS, LETTERS, AND PREPRINTS

- M. Bahrani, **M. Neuder**, S. M. Weinberg, “Adversarial procurement in blockchains,” (2026) *preprint*.
<https://arxiv.org/pdf/2605.05559>.
- P. Garimidi, **M. Neuder**, T. Roughgarden, “Tullock Contests in the Wild: Applications in Blockchains,” (2025) *ACM SIGecom Exchanges, Vol. 23, No. 1*.
https://www.sigecom.org/exchanges/volume_23/1/GARIMIDI.pdf.
- **M. Neuder**, D. J. Moroz, R. Rao, D. C. Parkes, “Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders,” (2020) *Workshop on Game Theory in Blockchain, Conference on Web and Internet Economics (WINE)*.
<https://arxiv.org/pdf/2102.02247>.

ACADEMIC TALKS

- **Beyond Winner-Take-All Procurement Auctions**
↳ Financial Cryptography and Data Security, *FC 2026*. [slides](#).
- **Selfish mining under general stochastic rewards**
↳ Advances in Financial Technology, *AFT 2025*. [recording](#). [slides](#).
- **Optimizing Exit Queues for Proof-of-Stake Blockchains**
↳ Advances in Financial Technology, *AFT 2024*. [slides](#).
- **Low-cost attacks on Ethereum 2.0 by sub-1/3 stakeholders**
↳ Workshop on Game Theory in Blockchain, *WINE 2020*. [recording](#). [slides](#).
- **Defending Against Malicious Reorgs in Tezos**
↳ ACM Advances in Financial Technology, *AFT 2020*. [recording](#). [slides](#).
- **Selfish Behavior in the Tezos PoS Protocol**
↳ Cryptoeconomic Systems Conference, *CES 2020*. [recording](#). [slides](#).
- **Animal Tracking using Deep Learning**
↳ Santa Fe Institute REU Presentation, *SFI 2018*. [recording](#).

AWARDS & FUNDING

2025-2030 Ethereum Foundation PhD Fellowship (award FY25-2276; \$200,000).

2020 Computer Science Discovery Learning Award (University of Colorado).
2019 Sieglinde Talbott Haller Scholarship in Mathematics (University of Colorado).
2019 Honorable Mention: Computing Research Assoc. Outstanding Undergraduate Researcher. [link.](#)
2017 Phi Beta Kappa (University of Colorado). [link.](#)
2015-2020 President Joseph A. Sewall Esteemed Scholar Award (University of Colorado). [link.](#)
2015-2020 Dean's List (University of Colorado). [link.](#)

PROFESSIONAL SERVICE

Columbia CryptoEconomics Workshop (CCE). ↳ 2023, 2024, 2025	Organizing Committee. link.
ACM Conference on Economics and Computation (EC). ↳ 2026	Program Committee. link.
Designing Defi (D^2) Conference. ↳ 2026	Program Committee. link.
The Latest in DeFi Research (TLDR) Conference. ↳ 2024, 2025	Program Committee. link.
Crypto & Blockchain Economics Research Forum (CBER) Conference. ↳ 2025, 2026	Program Committee. link.
Workshop on Decentralized Finance (DeFi) at Finanacial Cryptography. ↳ 2026	Program Committee. link.
Princeton Blockchain Special Interest Group (SIG). ↳ AY 2025-2026	Organizer.
Symposium On Discrete Algorithms (SODA). ↳ 2026	External Reviewer.
ACM Symposium on Principles of Distributed Computing (PODC). ↳ 2026	External Reviewer.
Crypto Academic Summer School at Edge City. ↳ 2024	Organizing Committee. link.
dYdX MEV Committee. ↳ 2023	Committee Member. link.

BLOCKCHAIN ARTICLES (NON-PEER REVIEWED) (BY TOPIC)

Proof-of-Stake

ETH, the asset

- My (e)thesis: settlement, data availability, execution – in that order. [link.](#) [tweet.](#)
- Issuance Issues – Tertiary Treatise. [link.](#) [tweet.](#)
- Issuance Issues – Subsequent Soliloquy. [link.](#) [tweet.](#)
- Issuance Issues – Initial Issue. [link.](#) [tweet.](#)

Ethereum Consensus

- Concurrent Block Proposers in Ethereum, *with Max Resnick.* [link.](#) [tweet.](#)
- Rollup-Centric Roadmap (2024 version), *with Alex Stokes.* [link.](#) [tweet.](#)
- A set-theoretic view of Ethereum coteries. [link.](#) [tweet.](#)
- Time, slots, and the ordering of events in Ethereum, *with Georgios Konstantopoulos.* [link.](#) [tweet.](#)

Increase the MAX_EFFECTIVE_BALANCE

- EIP-7251: Increase the MAX_EFFECTIVE_BALANCE. [link.](#)
- A modest proposal, *with Francesco D'Amato, Aditya Asgaonkar, Justin Drake.* [link.](#) [tweet.](#)
- Slashing penalty analysis; EIP-7251, *with Barnabe Monot.* [link.](#) [tweet.](#)
- Validator consolidation in EIP-7251, *with Francesco D'Amato, Mikhail Kalinin.* [link.](#) [tweet.](#)
- FAQ on EIP-7251, *with Francesco D'Amato, Mikhail Kalinin, dAppLion.* [link.](#) [tweet.](#)

Exit/Withdrawal queues

- EIP-7922: Dynamic exit queue rate limit, *with Mikhail, Mallesh.* [link.](#)
- Adding flexibility to Ethereum's exit queue, *with Mikhail Kalinin, Mallesh Pai.* [link.](#) [tweet.](#)

- ELI5: Ethereum Validator Exits, *with Mallesh Pai*. [link](#). [tweet](#).

Liquid staking & restaking

- The Risks of LRTs, *with Tarun Chitra*. [link](#). [tweet](#).
- Musings on “two-tiered” staking, a native Liquid Staking Token design. [link](#). [tweet](#).
- Magnitude and direction of Lido attack vectors. [link](#). [tweet](#).

Data availability/blobs

- Blob gossip and validation before and after PeerDAS. [link](#). [tweet](#).
- On the future of the blob mempool, *with Julian Ma*. [link](#). [tweet](#).

Maximal Extractable Value (MEV)

Enshrined Proposer-Builder Separation (ePBS)

- Why enshrine Proposer-Builder Separation, *with Justin Drake*. [link](#). [tweet](#).
- Payload-timeliness committee (PTC), *with Francesco D’Amato*. [link](#). [tweet](#).
- Equivocation attacks in mev-boost and ePBS, *with Francesco D’Amato*. [link](#). [tweet](#).
- Relays in a post-ePBS world, *with Jon, Hasu, Tomasz, Chris, Toni*. [link](#). [tweet](#).
- ePBS – the infinite buffet. [link](#). [tweet](#).
- Consider the ePBS. [link](#). [tweet](#).
- Decoupling ExecutionPayload validity from Data Availability. [link](#). [tweet](#).

Censorship resistance

- EIP-7547: Inclusion lists. [link](#).
- No free lunch – a new inclusion list design, *with Vitalik Buterin*. [link](#). [tweet](#).
- Unconditional inclusion lists, *with Toni Wahrstatter*. [link](#). [tweet](#).
- Resistance is not futile; CR in mev-boost. [link](#). [tweet](#).
- Inclusion lists: execution, consensus, & engine spec overview. [link](#). [tweet](#).
- Inclusion Lists PoC Specification. [link](#). [tweet](#).
- The Case for ILECTRA. [link](#). [tweet](#).

Relays

- Optimistic Relay Proposal, *with Justin Drake*. [link](#).
- An optimistic weekend. [link](#). [tweet](#).
- Towards enshrined PBS – an optimistic roadmap. [link](#).
- Optimistic relays and where to find them, *with Ankit Chiplunkar*. [link](#). [tweet](#).
- Bid cancellations considered harmful, *with Thomas Thiery*. [link](#). [tweet](#).

Procurement / zkEVM for Ethereum

- A native zkEVM scales bandwidth, not just execution [link](#). [tweet](#).
- Capacity Oracles, *with Maryam Bahrani*. [link](#). [tweet](#).
- On Ethereum Prover Market Design, *with Maryam Bahrani*. [link](#). [tweet](#).
- On incentivizing anonymous participation, *with Maryam Bahrani*. [link](#). [tweet](#).

Enshrined auction mechanisms

- Mechan-stein (alt. Franken-ism). [link](#). [tweet](#).
- On block-space distribution mechanisms, *with Pranav Garimidi, Tim Roughgarden*. [link](#). [tweet](#).
- Execution Tickets, *with Justin Drake*. [link](#). [tweet](#).
- How I learned to stop worrying and love mev-burn, *with Justin Drake, Toni Wahrstatter*. [link](#). [tweet](#).

Miscellanea

- Application-controlled execution: A case study on cancel prioritization, *with Maryam Bahrani*. [link](#). [tweet](#).
- Overnight reverse repurchases and the Fed’s balance sheet. [link](#). [tweet](#).
- Timing Games: Implications and Possible Mitigations, *with Caspar Schwarz-Schilling*. [link](#). [tweet](#).

BLOCKCHAIN PRESENTATIONS, PODCASTS, PANELS (REVERSE CHRONOLOGICAL)

Presentations

- How to get untrusted people to work for you – *Princeton DeCenter Spring Conference (April 2026)*. [slides](#).
- Blockchain governance in practice – *EGR 373 (Princeton) Guest talk. March 2026*. [slides](#).

- Selfish Mining under General Stochastic Rewards – *IC3 Winter Retreat. January 2026.* [slides.](#)
- Adversarial procurement – *Columbia CryptoEconomics (CCE). December 2025.* [recording.](#) [slides.](#)
- First Principles of Blockchains – *World Bank Session. November 2025.* [slides.](#)

EthCC[8] (July 2025) – Cannes, France

- ↳ Blobs and the future of the blob mempool – *EthCC main event.* [recording.](#) [slides.](#)
- ↳ Blob futures and the blob mempool – *Blockspace Futures Day.* [recording.](#) [slides.](#)
- Revisiting Ethereum’s Exit Queue – *Beam Chain Call #4. April 2025.* [recording.](#) [slides.](#)
- Validator incentives – *Oxford-Harvard Conference on DeFi. January 2025.* [slides.](#)

DEVCON 7 (November 2024) – Bangkok, Thailand

- ↳ ETH is permissionless money – *DEVCON Main stage.* [recording.](#) [slides.](#)
- ↳ The ticker is ETH – *Bankless Summit.* [recording.](#) [slides.](#)
- Neutrality in Ethereum – *Harvard Club of New York City. October 2024.* [slides.](#)
- Separating MEV and Staking Rewards – *CBER Webinar. September 2024.* [recording.](#) [slides.](#)
- More slides about Ethereum Blockspace – *Scroll Protocol Symposium. August 2024.* [recording.](#) [slides.](#)
- More slides about PBS – *MEV Workshop, Stanford Blockchain Conf. August 2024.* [recording.](#) [slides.](#)
- Postmodern Staking – *9th Annual IC3 Blockchain Camp. June 2024.* [slides.](#)
- Is restaking the new staking? – *CBER Forum Annual Conference. May 2024.* [recording.](#) [slides.](#)
- State of the Ethereum Union (1/N perspective) – *ETHBoston. April 2024.* [slides.](#)

ETHDenver (March 2024) – Denver, USA

- ↳ Execution Tickets – *Beyond the Block (Titan Builder & Hashkey Capital).* [recording.](#) [slides.](#)
- ↳ The Risks of LRTs – *Research Day (SevenX Ventures).* [recording.](#) [slides.](#)
- No Free Lunch – *Columbia CryptoEconomics Working Session. December 2023.* [recording.](#) [slides.](#)
- Enshrining PBS – *Center for Digital Finance & Technologies. December 2023.* [slides.](#)

DevConnect (November 2023) – Istanbul, Turkey

- ↳ Increase the MAX_EFFECTIVE_BALANCE – *EthStaker Staking Gathering.* [recording.](#) [slides.](#)
- ↳ A set theoretic view of Ethereum coteries – *LidoConnect.* [recording.](#) [slides.](#)
- Why it’s hard to enshrine PBS – *Archetype MEV Lunch. November 2023.* [recording.](#) [slides.](#)

EthCC[6] (July 2023) – Paris, France

- ↳ Towards Enshrined Proposer-Builder Separation – *EthCC main event.* [recording.](#) [slides.](#)
- ↳ Ethereum PBS R&D Roadmap – *Modular Summit.* [recording.](#) [slides.](#)
- ↳ Increase the MAX_EFFECTIVE_BALANCE – *Kiln Rendez-vous.* [recording.](#) [slides.](#)
- Reorgs in PoS – *MEV Roast; Reorg Edition. August 2021.* [recording.](#) [slides.](#)

Podcasts

- Pectra Explained – *Coinbase Institutional Webinar. May 2025.* [recording.](#)
- ETH is Permissionless Money – *Cryptoria. December 2024.* [English.](#) [Chinese.](#)
- Ethereum’s North Star – *The Gwart Show. October 2024.* [recording.](#)
- The Future of Ethereum: Is This The Right Track? – *Bankless. October 2024.* [recording.](#)
- The Ethereum Roadmap is NOT Off Track! – *Bankless. September 2024.* [recording.](#)
- ETH Insights: Discussing MaxEB – *Coinbase Webinar. April 2024.* [recording.](#) [article.](#)
- PeepAnEIP: EIP-7547 – *Ethereum Cat Herders. April 2024.* [recording.](#)
- Endgame 2.0: A Guide to Vitalik’s Ethereum Roadmap – *Bankless. February 2024.* [recording.](#)
- We’re Pretty Sure Mike and Max Can Fix MEV – *The Gwart Show. February 2024.* [recording.](#)
- PeepAnEIP: EIP-7251 – *Ethereum Cat Herders. February 2024.* [recording.](#)
- Eigenlayer In 2024 (co-host) – *Bankless. December 2023.* [recording.](#)
- An Incomplete Guide to PBS – *Uncommon Core 2. September 2023.* [recording.](#)

Panels

- Decentralization and pragmatism – *Princeton DeCenter Spring Conference. April 2026.*
- Restaking Panel – *Staking Rewards: Staking Summit. November 2024.* [recording.](#)

- Decentralization Panel – *Espresso: Beyond the Baselayer*. March 2024.
- Decentralize or Bust – *bloXroute & ETHStaker*. March 2024.
- Alignment Panel – *EignenLayer Restaking Summit*. November 2023.
- PBS & Beyond – *MEVDay Paris*. July 2023.

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OPEN SOURCE CONTRIBUTIONS

EIP-7547: Inclusion Lists (python) (repo)

- EIP (Ethereum Improvement Proposal).
- Spec Overview (Consensus, Execution, and Engine API specifications).
- Proof-of-Concept Specification.
- Compilation of Related Work.

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EIP-7251: Increase the MAX_EFFECTIVE_BALANCE (python) (repo)

- EIP (Ethereum Improvement Proposal).
- Minimal Spec Change (Consensus specifications).
- Compilation of Related Work.

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MEV-boost Relay: Optimistic Processing (golang) (repo)

- Main pull request & design documentation.
- Header-only parsing optimization.
- Pull request list.
- Proposal.
- Builder Onboarding.

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Erigon (Ethereum Execution Layer Client) (golang) (repo)

- Refactored req/resp domain network encoding.
- Consensus spec implementation.
- Pull request list.

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Prysm (Ethereum Consensus Layer Client) (golang) (repo)

- Keymanager code-health refactor.
- Pull request list.

link.
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STRUMPACK (High-performance matrix math package) (C++) (repo)

- Mixed-precision (float64 & float32) iterative refinement.
- Pull request list.
- Write-up.

link.
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OptRBC (Optimal solutions in 2D Rayleigh-Benard Convection) (FORTHAN) (repo)

- Multiprocessor implementation using [openmp](#).
- Write-up.

link.
link.

Image quality convolutional neural networks. (python) (repo)

- Write-up.
- Example network.

link.
link.

MISCELLANEA: FUN WRITING & SOFTWARE

Writing

- Tattoo probabilistic analysis.
- Solo-staking rig.
- NBA draft probabilistic analysis.
- Shakespeare Zipf-ian analysis.

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Software

- Monte Carlo numerical demonstrations. (python)
- Connect Four PyQT application. (python)
- Terminal implementation of games. (perl)
- Rubik's cube solver. (C++)

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